

Association between CTAs and survival - Fig1 - FigS1

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15-01-2026

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Load libraries

```
library(dplyr)
library(tidyr)
library(ComplexHeatmap)
library(circlize)
library(colorRamp2)
library(RColorBrewer)
library(survival)
library(forestplot)
library(survminer)
library(ggplot2)
library(ggrepel)
library(plotly)
library(ggsurvfit)
library(gridExtra)
library(clusterProfiler)
library(org.Hs.eg.db)
```

Functions

```
# Function to apply coxph model per columns
apply_coxph_model <- function(df) {
  # Empty df
  results <- data.frame()

  # Loop on each columns to apply coxph
  for (col in colnames(df)[3:ncol(df)]) {
    model <- coxph(Surv(OS.delay, OS.event) ~ df[[col]],
      data = df)

    # Extract results
    exp_coef <- summary(model)$coefficients[, "exp(coef)"]
    lower_ci <- summary(model)$conf.int[, "lower .95"]
    upper_ci <- summary(model)$conf.int[, "upper .95"]
    p_value <- summary(model)$coefficients[, "Pr(>|z|)"]
    results <- rbind(results, data.frame(Variable = col,
      HR = exp_coef, LowerCI = lower_ci, UpperCI = upper_ci,
      Pvalue = p_value))
  }
  return(results)
}

# Function to generate forest plot from coxph analysis from
# the function apply_coxph_model
generate_forestplot <- function(data, Type) {
  if (Type == TRUE) {
    # Order by Signature and p-value
    data <- data[order(data$Signature, data$Pvalue), ]
    data$Pvalue <- sprintf("%.4f", data$Pvalue)
  }
}
```

```

    data %>%
      forestplot(labeltext = c(Signature, Variable, Pvalue),
                mean = HR, lower = LowerCI, upper = UpperCI,
                grid = TRUE, zero = 1, col = fpColors(box = "black",
                line = "black"), hrzl_lines = TRUE, title = "Forest Plot for Cox Model",
                txt_gp = fpTxtGp(label = gpar(fontsize = 8)),
                ci.vertices = TRUE, boxsize = 0.5) %>%
      fp_set_zebra_style("#EFEFEF") %>%
      fp_add_header(Signature = c("", "Cell type"), Variable = c("",
        "Genes"), Pvalue = c("", "p-value"))
  } else {
    # Order by p-value
    data <- data[order(data$Pvalue), ]
    data$Pvalue <- sprintf("%.4f", data$Pvalue)
    data %>%
      forestplot(labeltext = c(Variable, Pvalue), mean = HR,
                lower = LowerCI, upper = UpperCI, grid = TRUE,
                zero = 1, col = fpColors(box = "black", line = "black"),
                hrzl_lines = TRUE, title = "Forest Plot for Cox Model",
                txt_gp = fpTxtGp(label = gpar(fontsize = 6)),
                ci.vertices = TRUE, boxsize = 0.3) %>%
      fp_set_zebra_style("#EFEFEF") %>%
      fp_add_header(Variable = c("", "Genes"), Pvalue = c("",
        "p-value"))
  }
}

# Function to calculate Z scores
calculate_z_scores <- function(df_input, col) {
  # Exclude col PROBEID SYMBOL and CTA
  data_values <- df_input[, -c(col)]

  # Calculate Z-scores
  z_scores_row <- t(scale(t(data_values)))

  # Add columns
  df_z_scores <- cbind(df_input[, c(col)], z_scores_row)

  # Return df
  return(df_z_scores)
}

```

Load data

```

l_cta <- read.table("../data/CTA_list_clean.txt", header = F)$V1
l_cta_validated <- read.table("../data/CTA_validated_list.txt",
  sep = "\t", header = FALSE)$V1

# Matrix with intensities and information on immune
# signature and CTA
df_CTA_immune_whole_clean_avg_102 <- read.table("../results/whole_gene_int_CTA_sign_imm_clean.tsv",

```

```

sep = "\t", header = TRUE, check.names = FALSE, row.names = 1)

# Metadata
df_metadata <- read.table("../results/metadata.tsv", sep = "\t",
  header = TRUE, check.names = FALSE, dec = ",")

# Conventional CHS metadata
df_metadata_conv <- df_metadata[df_metadata$Histology != "N/A" &
  df_metadata$Histology != "benign" & df_metadata$Histology !=
  "dedifferentiated", ]
df_metadata_surv_conv <- df_metadata_conv[, c("Patient", "OS.delay",
  "OS.event")]
df_metadata_surv_conv <- na.omit(df_metadata_surv_conv)

# Matrix for conventional patients with survival metadata
df_CTA_immune_whole_clean_avg_63 <- df_CTA_immune_whole_clean_avg_102[,
  c("Signature", "CTA", colnames(df_CTA_immune_whole_clean_avg_102)[colnames(df_CTA_immune_whole_clean_avg_102) != "Signature" & "CTA",
  df_metadata_surv_conv$Patient])]
df_z_scores_63 <- calculate_z_scores(df_CTA_immune_whole_clean_avg_63,
  c(1, 2))

# Consensus
df_hpa_raw <- read.table("../data/rna_tissue_consensus.tsv",
  sep = "\t", header = T)
df_hpa_cta <- df_hpa_raw[df_hpa_raw$Gene.name %in% l_cta, ]
df_hpa_cta <- as.data.frame(df_hpa_cta %>%
  pivot_wider(id_cols = Tissue, names_from = Gene.name, values_from = nTPM,
  values_fn = dplyr::first, values_fill = 0))
rownames(df_hpa_cta) <- df_hpa_cta$Tissue
df_hpa_cta <- df_hpa_cta[, -1]

```

I. CTA expression in healthy tissues

```

# Log10 to scale the data
df_log10 <- log10(df_hpa_cta + 1)

# CTA annotation
df_cta_validated <- df_log10[, colnames(df_log10) %in% l_cta_validated]
df_cta_putatives <- df_log10[, !colnames(df_log10) %in% l_cta_validated]

# Create heatmaps
colors = colorRamp2(c(min(df_log10), max(df_log10)), c("black",
  "red"))
ht_validated <- Heatmap(as.matrix(df_cta_validated), cluster_rows = FALSE,
  cluster_columns = FALSE, col = colors, border = NA, show_column_names = FALSE,
  show_row_names = TRUE, column_title = "Validated CTAs", row_names_gp = gpar(fontsize = 10),
  heatmap_legend_param = list(title = "Gene expression (Log10 TPM)",
  rect_gp = gpar(col = NA))

ht_putatives <- Heatmap(as.matrix(df_cta_putatives), cluster_rows = FALSE,
  cluster_columns = FALSE, col = colors, show_column_names = FALSE,

```

```
show_row_names = TRUE, column_title = "Putative CTAs", row_names_gp = gpar(fontsize = 10),  
rect_gp = gpar(col = NA))  
  
ht_validated + ht_putatives
```

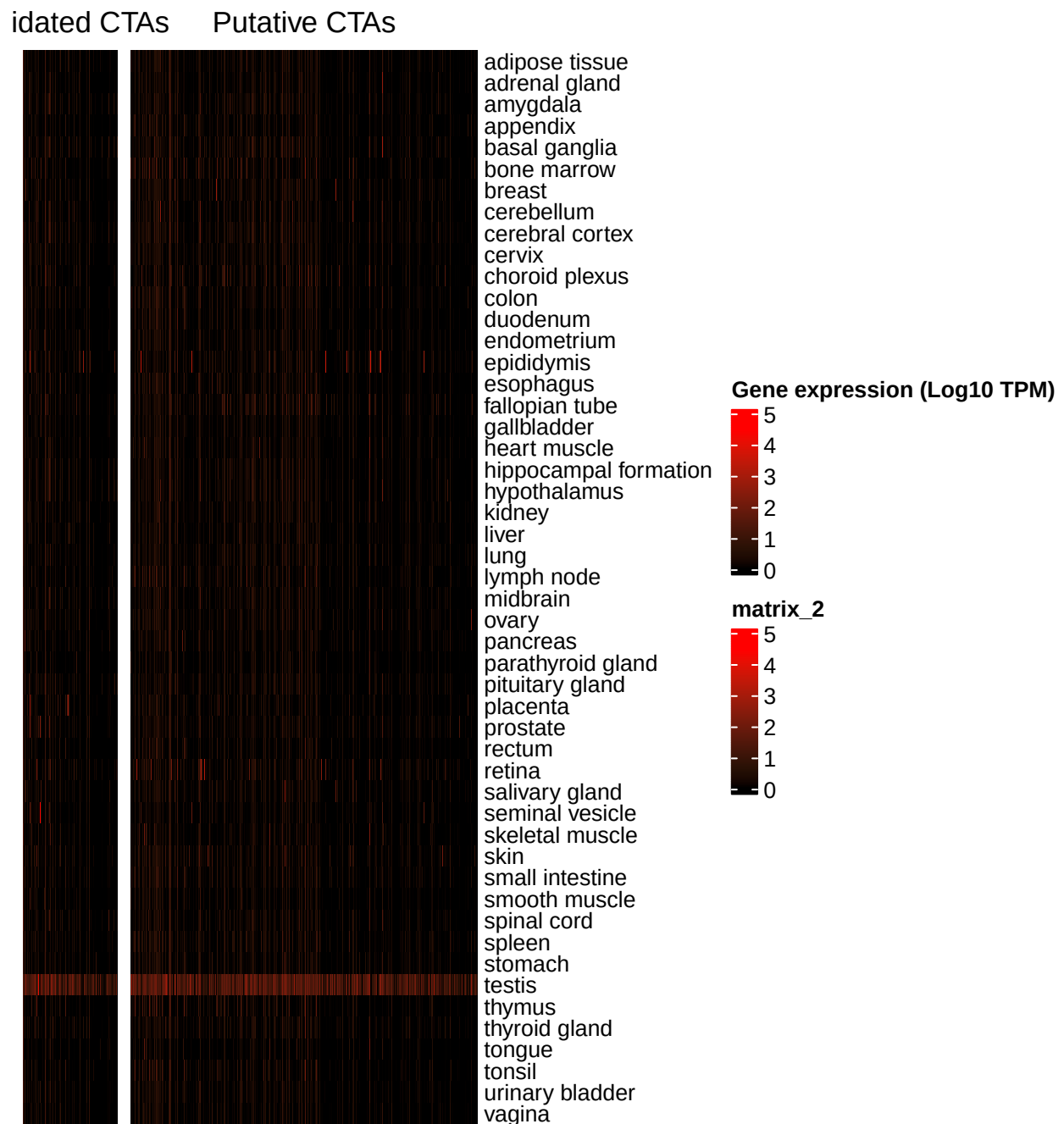


Figure 1: Heatmap of CTAs expression in normal tissues (HPA-GTEx consensus)

II. Survival analyses

```
# CTA data
df_expr_cta_63 <- subset(df_z_scores_63, CTA == "CTA")
df_expr_cta_63 <- df_expr_cta_63[, !colnames(df_expr_cta_63) %in%
  c("Signature", "CTA", "SYMBOL")]
df_expr_cta_63 <- as.data.frame(t(df_expr_cta_63))
df_expr_cta_63$Patient <- rownames(df_expr_cta_63)
df_expr_cta_63 <- merge(df_metadata_surv_conv, df_expr_cta_63,
  by = "Patient")
rownames(df_expr_cta_63) <- df_expr_cta_63$Patient
df_expr_cta_63 <- df_expr_cta_63[, -1]

# Apply coxph and generate forest plot with significant
# data
res_cta <- apply_coxph_model(df_expr_cta_63)
res_cta_signif <- res_cta[res_cta$Pvalue < 0.05, ]
res_cta_sorted <- res_cta_signif[order(res_cta_signif$HR, decreasing = TRUE),
  ]

# Select top 20 highest HR
top20_hr <- head(res_cta_sorted, 20)

# Select top 20 smallest HR
bottom20_hr <- head(res_cta_sorted[order(res_cta_sorted$HR, decreasing = FALSE),
  ], 20)

# Combine
res_cta_extremes <- rbind(top20_hr, bottom20_hr)

# Forestplot
generate_forestplot(res_cta_extremes, Type = FALSE)
```

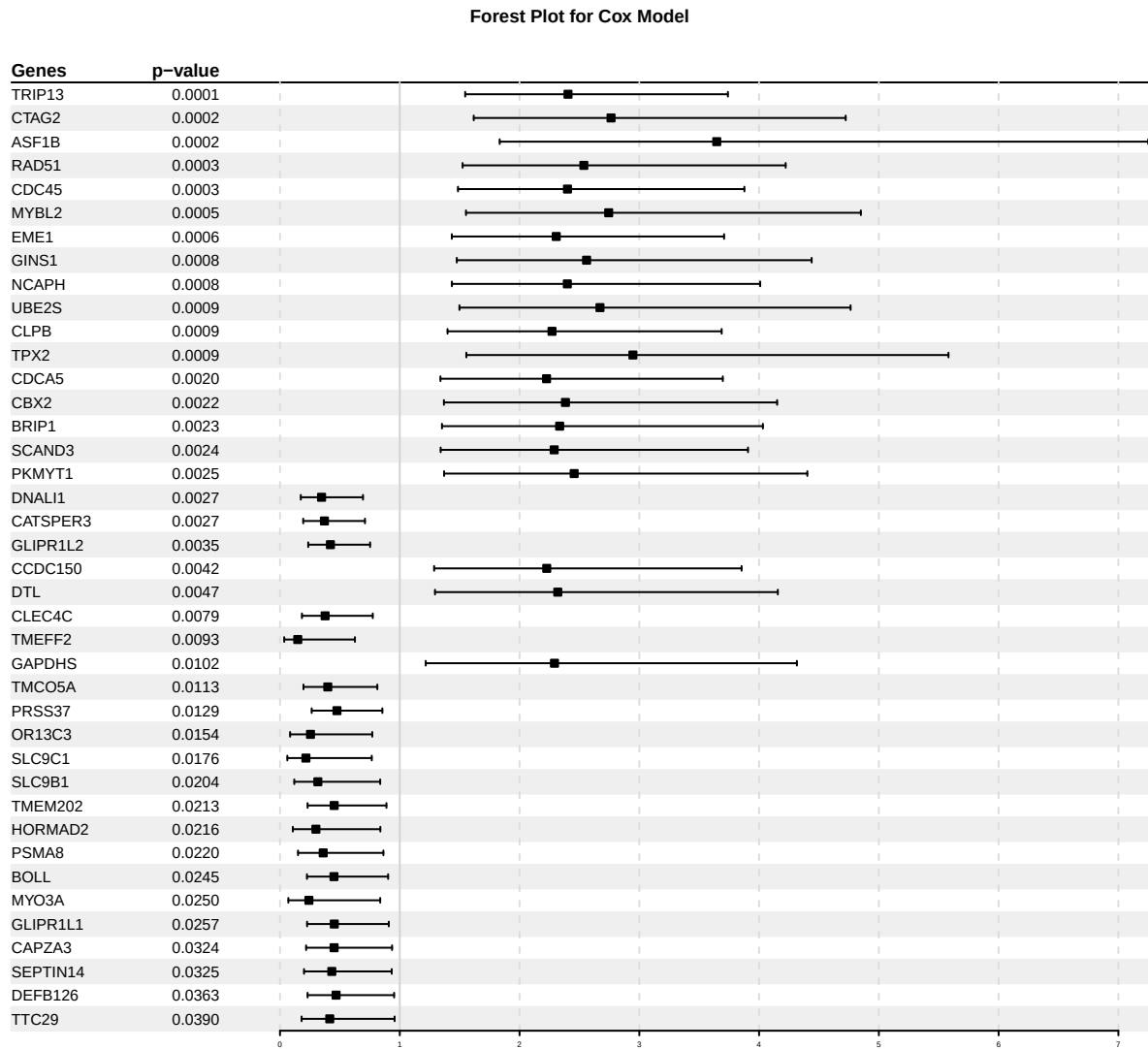


Figure 2: Forest plot for the 20 CTAs with the highest HR and the 20 with the lowest HR (n = 63)

III. Heatmap of CTAs significantly associated with survival

```
# List of CTA for corph analysis
l_CTA_conv <- res_cta_signif$Variable
data_cta_63 <- df_z_scores_63[rownames(df_z_scores_63) %in% l_CTA_conv,
  -c(1, 2)]

# Create the heatmap
colors <- colorRampPalette(c("blue", "white", "red"))(100)
heatmap_cta_signif_conv_63 <- Heatmap(as.matrix(data_cta_63),
  cluster_rows = TRUE, cluster_columns = TRUE, cluster_column_slices = TRUE,
  clustering_distance_columns = "euclidean", clustering_method_columns = "complete",
  show_column_dend = TRUE, col = colorRamp2(seq(-8, 8, length.out = 100),
    colors), border = NA, show_column_names = TRUE, show_row_names = TRUE,
  column_names_gp = gpar(fontsize = 4), row_names_gp = gpar(fontsize = 2),
  heatmap_legend_param = list(title = "Expression level (z_scores)"))
heatmap_cta_signif_conv_63 <- draw(heatmap_cta_signif_conv_63)
```

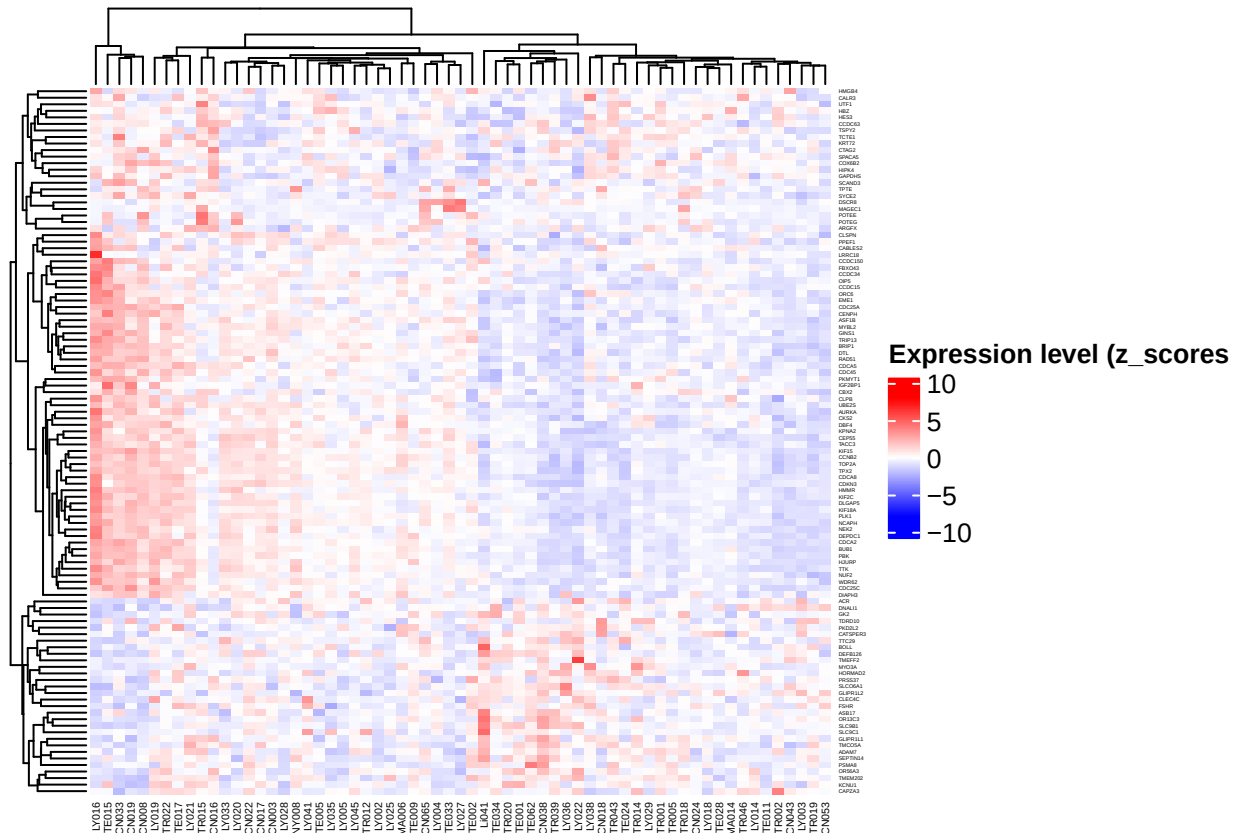


Figure 3: Heatmap of CTAs that impact survival (n = 63)

```

# Store clusters
col_dend <- column_dend(heatmap_cta_signif_conv_63)
row_dend <- row_dend(heatmap_cta_signif_conv_63)

# Cut in clusters
col_clusters <- cutree(as.hclust(col_dend), k = 3)
row_clusters <- cutree(as.hclust(row_dend), k = 3)

# Select colors
histology_colors <- c(G1 = "#9e9e9eff", G2 = "#625f6aff", G3 = "#000000ff")

# Heatmap
Heatmap(as.matrix(data_cta_63), cluster_rows = TRUE, cluster_columns = TRUE,
        cluster_column_slices = TRUE, column_split = col_clusters,
        row_split = row_clusters, clustering_distance_columns = "euclidean",
        clustering_method_columns = "complete", show_column_dend = TRUE,
        col = colorRamp2(seq(-8, 8, length.out = 100), colors), border = NA,
        show_column_names = F, show_row_names = F, heatmap_legend_param = list(title = "Z-scores of intensi
        top_annotation = columnAnnotation(Histology = df_metadata_conv[df_metadata_conv$Patient %in%
        colnames(data_cta_63), "Histology"], col = list(Histology = histology_colors)))

```

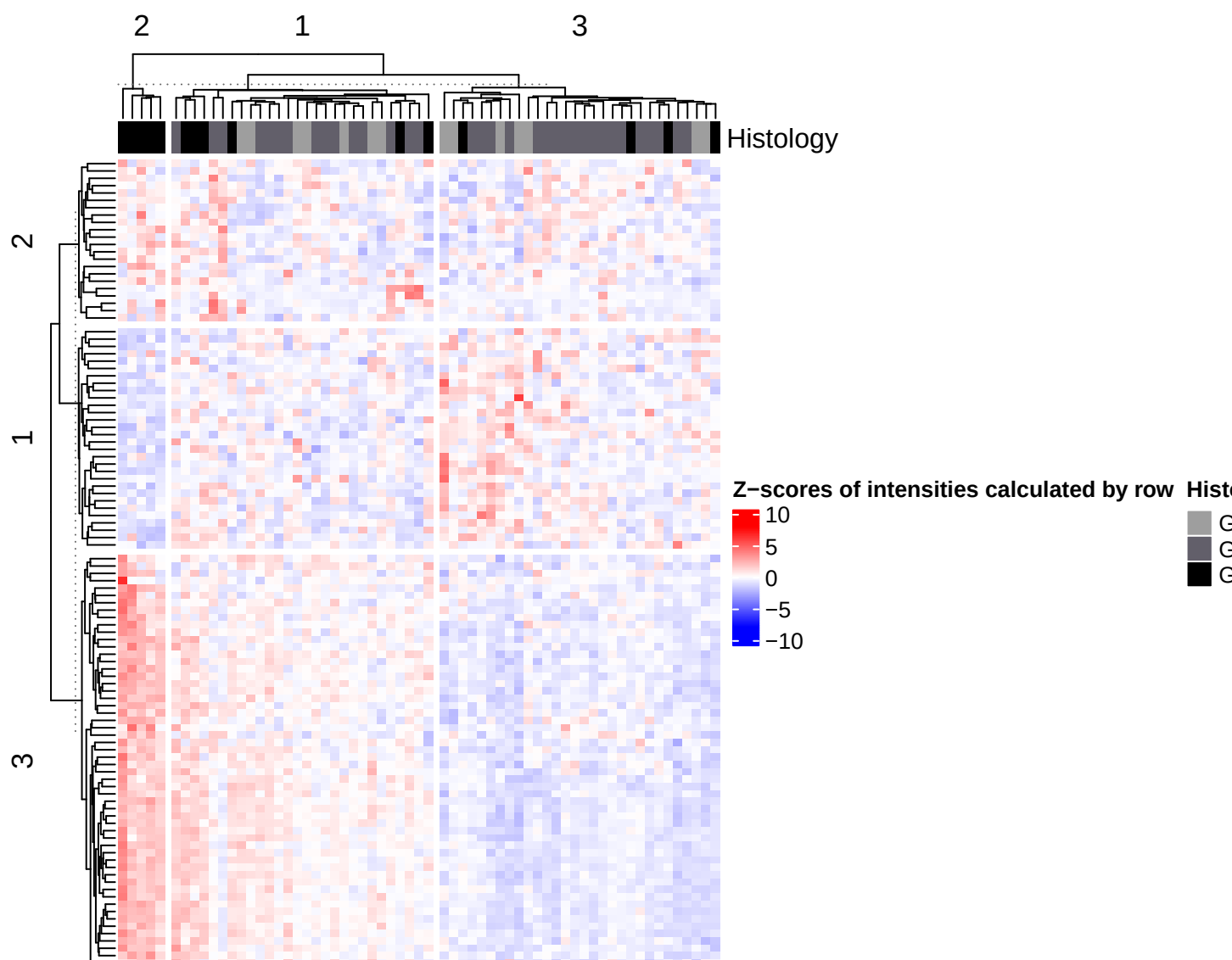
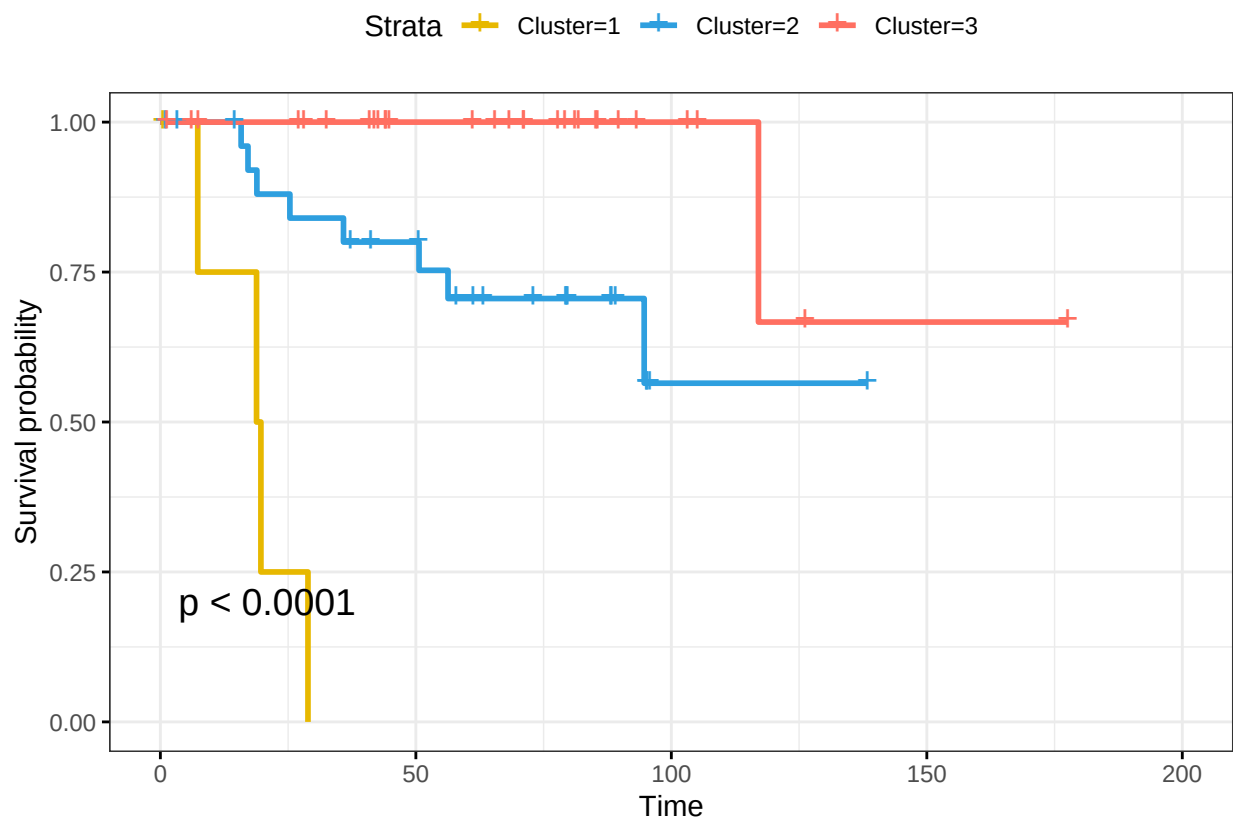


Figure 4: Significant CTAs heatmap with grade data (n = 63)

IV. Kaplan Meier plot

```
# Read clusters
col_clusters <- as.data.frame(col_clusters)
colnames(col_clusters) <- "Cluster"
col_clusters$Patient <- rownames(col_clusters)
col_clusters$Cluster[col_clusters$Cluster == "1"] <- "tmp"
col_clusters$Cluster[col_clusters$Cluster == "2"] <- "1"
col_clusters$Cluster[col_clusters$Cluster == "tmp"] <- "2"
df_km <- merge(col_clusters, df_metadata_surv_conv, by = "Patient")
rownames(df_km) <- df_km$Patient
km_plot <- ggsurvplot(survfit(Surv(OS.delay, OS.event) ~ Cluster,
  data = df_km), pval = TRUE, conf.int = FALSE, ggtheme = theme_bw(),
  palette = c("#E7B800", "#2E9FDF", "#FF6F61"))
km_plot$plot
```



V. CTAs expression in CHS and normal tissues

```
# Means consensus
df <- as.data.frame(t(df_hpa_cta[!rownames(df_hpa_cta) == "testis",
]))
df_mean_tissue <- as.data.frame(rowMeans(df))
colnames(df_mean_tissue) <- "Mean_expression_tissues"
df_mean_tissue$SYMBOL <- rownames(df_mean_tissue)

# Means CTA
df_int_cta <- subset(df_CTA_immune_whole_clean_avg_63, CTA ==
"CTA")
df_int_cta <- df_int_cta %>%
  dplyr::select(-Signature, -CTA)
df_mean_expr <- as.data.frame(rowMeans(df_int_cta))
colnames(df_mean_expr) <- "Intensity"
df_mean_expr$SYMBOL <- rownames(df_mean_expr)

# Merge
df_scatter <- merge(df_mean_expr, df_mean_tissue, by = "SYMBOL")
df_clust_cta <- as.data.frame(row_clusters)
colnames(df_clust_cta) <- "Cluster"
df_clust_cta$SYMBOL <- rownames(df_clust_cta)
df_clust_cta$Cluster[df_clust_cta$Cluster == "1"] <- "B"
df_clust_cta$Cluster[df_clust_cta$Cluster == "2"] <- "A"
df_clust_cta$Cluster[df_clust_cta$Cluster == "3"] <- "C"

# Write table write.table(df_clust_cta,
# '../results/clusters_cta_signif_conv_indiv.tsv', sep =
# '\t', row.names = FALSE, quote = FALSE)

df_scatter <- merge(df_scatter, df_clust_cta, by = "SYMBOL",
all.x = T)
df_scatter$Cluster <- as.character(df_scatter$Cluster)
df_scatter$Mean_expression_tissues[df_scatter$Mean_expression_tissues ==
0] <- 0.001
df_scatter$Mean_expression_tissues_log10 <- log10(df_scatter$Mean_expression_tissues)

# Adding HR data
df_hr <- res_cta_signif[, c("Variable", "HR")]
df_hr$SYMBOL <- df_hr$Variable
df_scatter_hr <- merge(df_scatter, df_hr, by = "SYMBOL", all.x = T)

# Add HR value to add point size
df_scatter_hr$HR <- ifelse(is.na(df_scatter_hr$HR), 0.1, df_scatter_hr$HR)

# Save df write.table(df_scatter_hr,
# '../results/table_scatter_63_complete.tsv', sep = '\t',
# quote = F)

# plot
p <- ggplot(df_scatter_hr, aes(x = Mean_expression_tissues, y = Intensity,
color = Cluster, size = HR, text = paste0("Gene: ", SYMBOL,
```

```

    "<br>Mean expression: ", round(Mean_expression_tissues,
      3), "<br>Intensity: ", round(Intensity, 3), "<br>HR: ",
      round(HR, 3), "<br>Cluster: ", Cluster))) + geom_point(alpha = 0.8) +
    geom_vline(xintercept = quantile(df_scatter_hr$Mean_expression_tissues,
      0.75), linetype = "dashed", color = "black") + geom_hline(yintercept = median(df_scatter_hr$Int
      linetype = "dashed", color = "black") + annotate("text",
    x = max(df_scatter_hr$Mean_expression_tissues), y = median(df_scatter_hr$Intensity) +
      0.3, label = "Median\nexpression", size = 2.5) + annotate("text",
    x = quantile(df_scatter_hr$Mean_expression_tissues, 0.75) +
      4, y = max(df_scatter_hr$Intensity) + 0.2, label = "75th percentile",
      size = 2.5) + scale_size_continuous(range = c(1, 4), breaks = c(min(df_scatter_hr[!is.na(df_scatter
      "HR"]), 1, max(df_scatter_hr$HR, na.rm = TRUE))), labels = ~round(.x,
      2)) + scale_color_manual(values = c(A = "#3498DB", B = "#2ECC71",
      C = "#E74C3C"), na.value = "#b0b0b0") + theme_minimal() +
    scale_x_log10(breaks = c(0.01, 1, 100), labels = c("0.01",
      "1", "100")) + labs(x = "CTAs expression in normal tissues (Log10 TPM mean)",
      y = "CTAs expression in CHS (mean intensities)", size = "Sized according\nto HR(only for\nuncolored p
      color = "CTAs cluster")

```

p

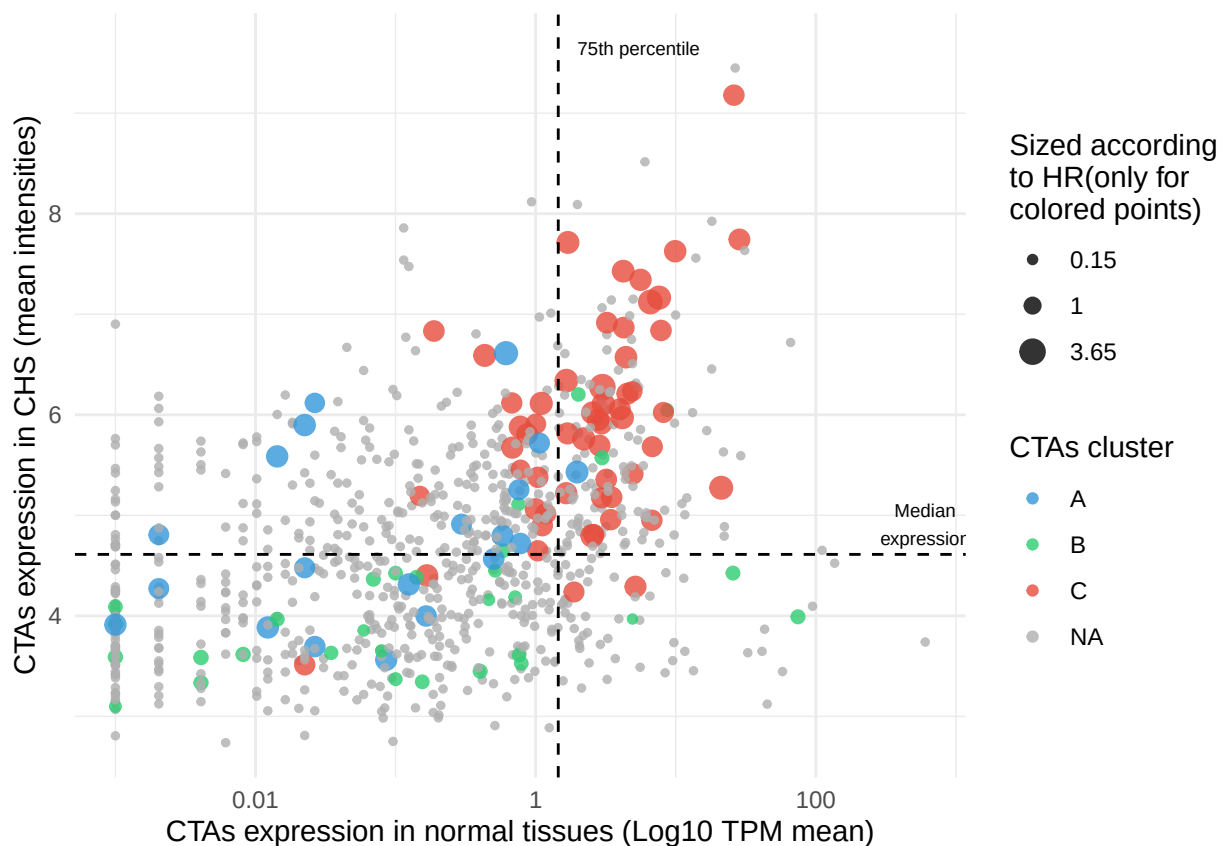


Figure 5: Scatter plot


```

df <- df_scatter_hr[df_scatter_hr$Intensity > median(df_scatter_hr$Intensity) &
  df_scatter_hr$Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), ]

# Genes to add label
genes <- c("SDK2", "TPTE", "UTF1", "HES3", "HMGB4", "DMRTC2",
  "DMBX1", "HHIPL2", "TEKT4", "CT45A1", "VCX2", "PCSK4", "TCTE1",
  "KIF15", "FHAD1", "SYCE2", "KIF18A", "CDCA2", "CDC25C", "COX6B2",
  "ACR", "HIPK4", "CCDC15", "EME1", "PPEF1", "DIAPH3", "CBX2",
  "AKR1E2", "PNMA5", "PRND", "CTAG2", "BRIP1", "SPATC1", "DBF4",
  "TMEM89", "POU4F1", "IGF2BP1", "PRAME", "CATSPER3", "CLSPN",
  "DEPDC1", "HBZ", "STAG3", "LRRC18", "MAGEA3", "MAGEA1", "MAGEC2",
  "FAM170B", "C2orf78", "PRPS1L1", "CT47B1", "POTEC", "DMP1",
  "C3orf22", "DNAJB8", "ACTL9")

# Plot
p <- ggplot(df, aes(x = Mean_expression_tissues, y = Intensity,
  color = Cluster, size = HR, text = paste0("Gene: ", SYMBOL,
    "<br>Mean expression: ", round(Mean_expression_tissues,
      3), "<br>Intensity: ", round(Intensity, 3), "<br>HR: ",
      round(HR, 3), "<br>Cluster: ", Cluster))) + geom_point(alpha = 0.8) +
  geom_hline(yintercept = median(df_scatter_hr$Intensity),
    linetype = "dashed", color = "black") + geom_vline(xintercept = quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), linetype = "dashed", color = "black") + scale_color_manual(values = c(A = "#3498DB",
    B = "#2ECC71", C = "#E74C3C"), na.value = "#b0b0b0") + theme_minimal() +
  scale_x_log10() + scale_size_continuous(range = c(1, 4),
    breaks = c(min(df_scatter_hr[!is.na(df_scatter_hr$Cluster),
      "HR"]), 1, max(df_scatter_hr$HR, na.rm = TRUE)), labels = ~round(.x,
      2)) + scale_color_manual(values = c(A = "#3498DB", B = "#2ECC71",
    C = "#E74C3C"), na.value = "#b0b0b0") + theme_minimal() +
  scale_x_log10() + geom_text_repel(data = df[df$SYMBOL %in%
    genes, ], aes(label = SYMBOL), size = 3, color = "black",
    max.overlaps = 20, box.padding = 0.4, segment.color = "black",
    segment.size = 0.3, min.segment.length = 0) + labs(x = "CTAs expression in normal tissues (Log10 TPM)",
    y = "CTAs expression in CHS (mean intensities)", size = "Sized according\nto HR(only for\ncolored points)",
    color = "CTAs cluster")

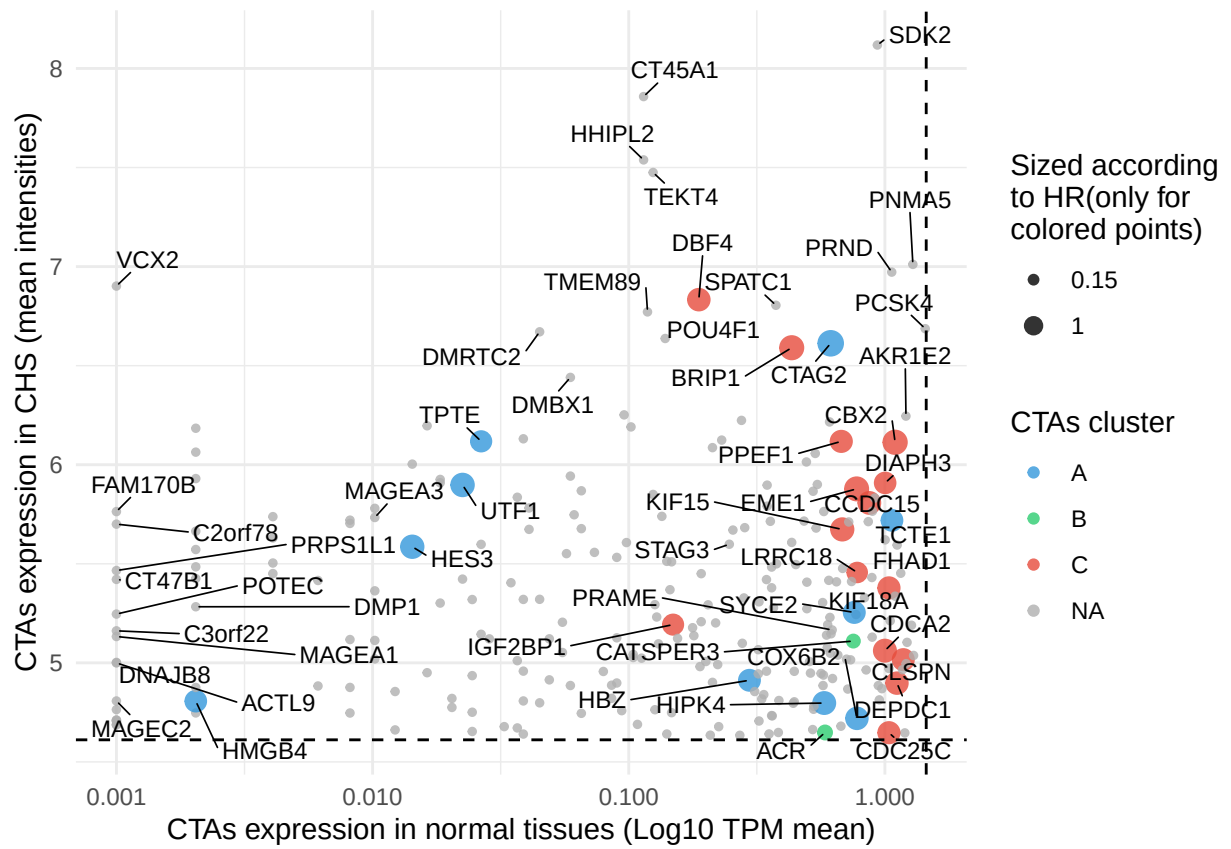
```

```

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.

```

```
p
```



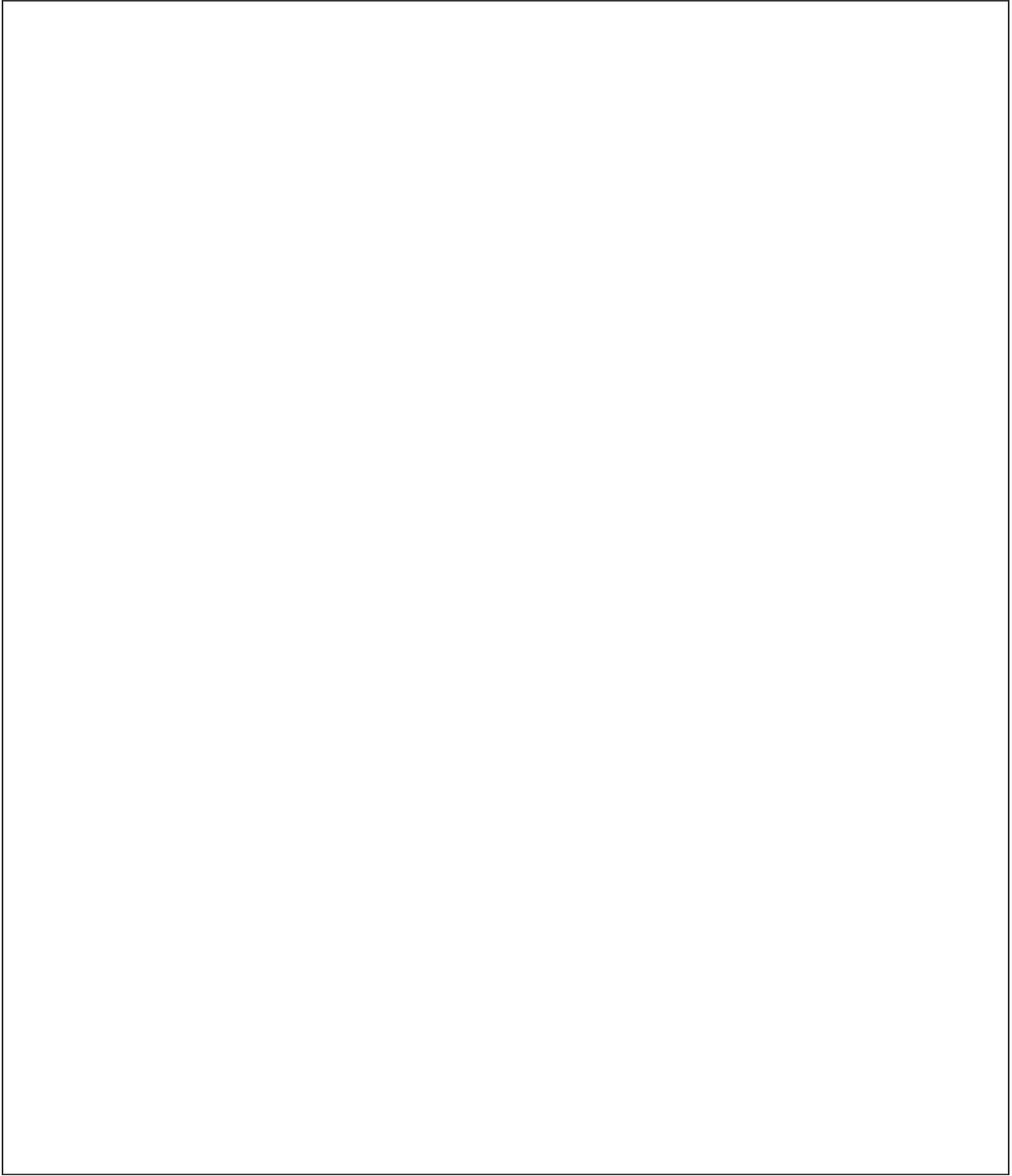
```
# ggplotly(p, tooltip = 'text')
```

VI. Gene ontology

```
cluster_a <- df_scatter_hr$SYMBOL[df_scatter_hr$Cluster == "A" &  
  !is.na(df_scatter_hr$Cluster)]  
cluster_b <- df_scatter_hr$SYMBOL[df_scatter_hr$Cluster == "B" &  
  !is.na(df_scatter_hr$Cluster)]  
cluster_c <- df_scatter_hr$SYMBOL[df_scatter_hr$Cluster == "C" &  
  !is.na(df_scatter_hr$Cluster)]
```

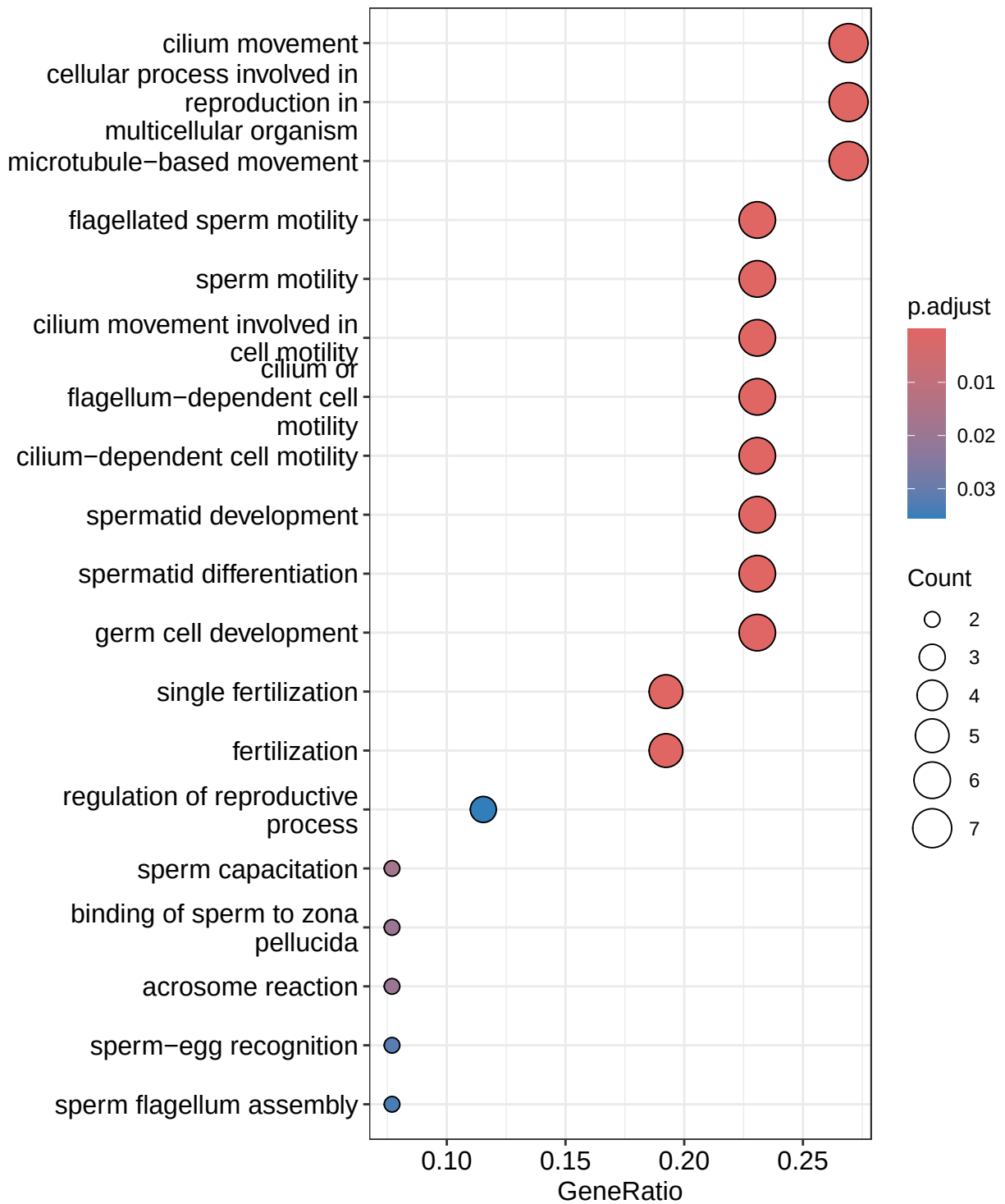
```
go_a <- enrichGO(gene = cluster_a, OrgDb = org.Hs.eg.db, keyType = "SYMBOL",  
  ont = "BP", pAdjustMethod = "BH", pvalueCutoff = 0.05)  
  
go_b <- enrichGO(gene = cluster_b, OrgDb = org.Hs.eg.db, keyType = "SYMBOL",  
  ont = "BP", pAdjustMethod = "BH", pvalueCutoff = 0.05)  
  
go_c <- enrichGO(gene = cluster_c, OrgDb = org.Hs.eg.db, keyType = "SYMBOL",  
  ont = "BP", pAdjustMethod = "BH", pvalueCutoff = 0.05)
```

```
dotplot(go_a, showCategory = 15)
```



GeneRatio

```
dotplot(go_b, showCategory = 20)
```



```
dotplot(go_c, showCategory = 20)
```

